HashMap<K,V> program :

```
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 package com.ravi.hash_map;

 import java.util.Collection;
 import java.util.HashMap;
 import java.util.Iterator;
 import java.util.Map;
 import java.util.Map.Entry;

 //BiConsumer  keySet()+loop  get()   getOrDefault()
 //containsKey()  containsValue()    entrySet().iterator
 public class HashMapDemo1
 {
      public static void main(String[] args)
      {
       HashMap<Integer, String> map = new HashMap<>();
       map.put(1, "Vanilla");
       map.put(2, "Butterscotch");
       map.put(3, "Chocolate");
       map.put(4, "Cotton Candy");

       IO.println("Map entries :"+map);

       IO.println("Using forEach() Method :");
       map.forEach((key, value)-> IO.println(key+" : "+value));

       for(Integer key : map.keySet())
       {
           IO.println("Key is :"+key);
       }

       Collection<String> values = map.values();
       IO.println("All the values are : ");
       IO.println(values);

       Integer key = 3;
       IO.println("Value of key "+ key +" is :"+map.get(key));

       key = 1;
       IO.println("Value of key "+ key +" is :"+map.getOrDefault(key,"Key is not available"));

       boolean isKeyAvailable = map.containsKey(2);
       IO.println("Is key available :"+isKeyAvailable);

       boolean isValueAvailable = map.containsValue("Butterscotch");
       IO.println("Is value available :"+isValueAvailable);

       IO.println("Iterating through Iterator :");

       Iterator<Entry<Integer, String>> iterator = map.entrySet().iterator();
       iterator.forEachRemaining(IO::println);

      }
 }
```

```
 package com.ravi.hash_map;

 import java.util.HashMap;
 import java.util.Map;

 class LoginService
 {
     private Map<String, String> users;

     public LoginService()
     {
         users = new HashMap<>();
     }

     public void register(String name, String password)
     {
         users.put(name, password);
         IO.println("User registration completed!!!");
     }

     public boolean isValidUser(String name, String password)  //ravi   1234
     {
         if(users.containsKey(name)  && users.get(name).equals(password))
         {
             return true;
         }
         else
         {
             return false;
         }
     }
 }




 public class HashMapDemo2
 {
     public static void main(String[] args)
     {
     LoginService login = new LoginService();
     login.register("Ravi", "1234");

     String name = IO.readln("Enter your username :");
     String pass = IO.readln("Enter your password :");

     if(login.isValidUser(name, pass))
     {
       IO.println("Hello "+name+" you are authenticated!!!");
     }
     else
     {
       System.err.println("Sorry!!! wrong username/password");
     }



     }
 }
```

```
package com.ravi.hash_map;
// putAll() keySet()  values() + Loose coupling

import java.util.Collection;
import java.util.HashMap;
import java.util.Set;

record Student(Integer id, String name)
{

}


public class HashMapDemo3
{
  public static void main(String[] args)
  {
       HashMap<Integer,String> newmap1 = new HashMap<>();
         HashMap<Integer,String> newmap2 = new HashMap<>();

         newmap1.put(1, "OCPJP");
         newmap1.put(2, "is");
         newmap1.put(3, "best");

         newmap2.put(4, "exam");
         newmap2.put(5, "by Oracle");

         IO.println("By using putAll() method :");
       newmap1.putAll(newmap2);

       IO.println(newmap1);


       Set<Integer> keySet = newmap1.keySet();
       IO.println("All the keys :"+keySet);

       Collection<String> values = newmap1.values();
       IO.println("All the values in map2 :"+values);


       HashMap<Student,String> studentMap = new HashMap<>();
       studentMap.put(new Student(111, "Scott"), "Hyderabad");
       studentMap.put(new Student(222, "Smith"), "Hyderabad");
       studentMap.put(new Student(333, "Allen"), "Hyderabad");
       studentMap.put(new Student(444, "Black"), "Hyderabad");

       HashMap<Student, String> duplicateMap = new HashMap<>(studentMap);
       duplicateMap.forEach((k,v)-> IO.println(k+" : "+v));

  }
}
```

```
package com.ravi.hash_map;

import java.util.HashMap;

record Employee(Integer empId, String empName)
{
}
public class HashMapDemo4
{
    public static void main(String[] args)
    {
        Employee e1 = new Employee(101,"Aryan");
        Employee e2 = new Employee(102,"Pooja");
        Employee e3 = new Employee(101,"Aryan");
        Employee e4 = e2;


        HashMap<Employee,String> hm = new HashMap<>();
        hm.put(e1,"Ameerpet");      //e1 =  Begumpet
        hm.put(e2,"S.R Nagar");     //e2 = Panjagutta
        hm.put(e3,"Begumpet");
        hm.put(e4,"Panjagutta");
        hm.forEach((k,v)-> System.out.println(k+" : "+v));
    }
}
```

```
 package com.ravi.hash_map;

 import java.util.HashMap;
 import java.util.Map;

 public class HashMapDemo5
 {
    public static void main(String[] args)
    {
      // Create a HashMap to store book titles and their availability (true =       available, false = borrowed)

      HashMap<String, Boolean> library = new HashMap<>();

      library.put("Core Java", true);
      library.put("Advanced Java", true);
      library.put("HTML", false);
      library.put("JavaScript", true);

      IO.println("Initial Library Status :");

      for(HashMap.Entry<String,Boolean> entry : library.entrySet())
      {
          IO.println(entry.getKey()+" : "+entry.getValue());
      }

     // Borrow a book
     String bookToBorrow = "Advanced Java";

     if(library.containsKey(bookToBorrow) && library.get(bookToBorrow))
     {
       library.put(bookToBorrow, false);
       IO.println(bookToBorrow+ " has borrowed successfully...");
     }
     else
     {
       IO.println(bookToBorrow+" book is not available for borrow");
     }

     String bookToReturn = "HTML";

     if(library.containsKey(bookToReturn) && !library.get(bookToReturn))
     {
       library.put(bookToReturn, true);

       IO.println(bookToReturn+" has returned by the borrower ");
     }
     else
     {
       IO.println(bookToReturn+" is not in the library");
     }


      library.forEach((k,v)-> System.out.println(k+" : "+v));


      // Check the availability of a book
      String bookToCheck = "JavaScript";

      if (library.containsKey(bookToCheck))
      {
        String availability = library.get(bookToCheck) ? "available" : "borrowed";
          IO.println(bookToCheck + " Book is " + availability + ".");
      }
      else
      {
          System.out.println(bookToCheck + " is not in the library.");
      }


      //Display the final library status

      System.out.println("Final Library Status:");
      for (Map.Entry<String, Boolean> entry : library.entrySet())
      {
          String status = entry.getValue() ? "Available" : "Borrowed";
          System.out.println("Book: " + entry.getKey() + ", Status: " + status);
      }


    }
 }
```

```
package com.ravi.hash_map;

import java.util.HashMap;
import java.util.Map;

//IQ Frequency count (count the Frequency of each word)
public class HashMapDemo6
{
    public static void main(String[] args)
    {
        Map<String,Integer> frequency = new HashMap<>();

        String text = "java is fun, java is powerful java is great";

        String []words = text.split(" ");

        for(String word : words)
        {
           frequency.put(word, frequency.getOrDefault(word, 0)+1);
        }

        IO.println(frequency);
    }

    /*java = 3
     * is = 3
     * fun, = 1
     *
     */
}
```